M.M.C.O.E.

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FAQ's

1) What is Multiple Inheritance?

→ It is an object-oriented mechanism where a single derived class inherits directly from more than one base class.

2) How is multiple inheritance different from multi-level inheritance?

→ Multiple inheritance involves one class inheriting from multiple bases simultaneously, whereas multilevel inheritance chains classes in a parent-child sequence (A → B → C).

3) Why are protected members useful in inheritance?

→ They allow derived classes to access base class members directly while keeping them hidden from outside code.

4) What is the effect of public inheritance?

→ It preserves the public & protected accessibility of base class members within the derived class, modelling an "is-a" relationship.

5) What ambiguity can occur in multiple inheritance, & how can it be resolved?

→ Ambiguity arises when multiple base classes share members with identical names; it can be resolved using scope resolution operator (::).

6) Write a C++ program to implement single inheritance.

→
```
#include<iostream>
using namespace std;
class Person
{
 public:
 string name;
 void getPersonData()
 {
   cout<<"Enter name: ";
   cin>>name;
 }
};
class Student :: public Person
{
 private:
  int rollno;
 public:
   void getStudentData(){
       getPersonData();
       cout<<"Enter roll number:";
       cin>>rollno;
   }
   void display()
{
 cout<<"Student Info";
 cout<<"Name :<<name<<endl;
 cout<<"Roll number:<<rollno<<endl;
}};
```

```
int main()
{
  Student s;
  s.getStudentData();
  s.display();
  return 0;
}
```

7) Write a C++ program to implement multilevel inheritance.

→

```
#include <iostream>
using namespace std;
class Student
{
  protected:
    int rollno.;
  public:
    void getNumber()
    {
     cout << "Enter roll number :";
     cin >> rollno.;
    }
};

class Test : public Student {
{
  protected:
    float sub1, sub2;
  public:
    void getMarks()
```

```
    {
     cout<<"Enter marks of two subjects :";
     cin >> sub1 >> sub2;
    }
};

class Result : public Test
{
 public :
    void display()
    {
     float total = sub1 + sub2;
     cout << "In Result " <<endl;
     cout << " Roll number : "<<rollno.<<endl;
     cout<<" Total marks : "<< total <<endl;
    }
};

int main ()
{
 Result studentResult;
 studentResult.getNumber();
 studentResult.getMarks();
 studentResult.display();
 return 0;
}
```